

FLY TO THE MOON
The Chandrayaan-3 launch in July , 2023

PRIVATE MISSION

Backing research and private participation, the Budget underlines the key role of space and atomic energy

BY **AMARNATH K. MENON**

IN A FEW FIGURES AND WORDS, THE UNION BUDGET CAPTURED TWO REALITIES. One, how vital India's state-steered space and atomic energy sectors are to the vision of Viksit Bharat. And two, how the idea of opening them up to the private sector is seen as critical to their evolution. Finance minister Nirmala Sitharaman has allocated Rs 13,042.75 crore to the space sector, up from the Rs 12,543.91 crore allocated in 2023-24. She also announced the setting up of a venture capital fund worth Rs 1,000 crore to push development in space technology. The capital, which Sitharaman said will expand the space economy by five times in the next 10 years, is likely to aid more than 180 government-recognised start-ups working in the field of space technology. Talking about the atomic energy sector, the finance minister announced that the Centre will work with private players to develop small modular reactors (SMRs), which will help India expand its nuclear energy generation. "Nuclear energy is expected to form a significant part of the energy mix for Viksit Bharat," said Sitharaman.

SMRs are reactors with a capacity of 300 MW or less per unit, which can be built in factories, unlike conventional nuclear reactors that are built on-site. The smaller size and factory production not only helps save cost but also construction time, with SMRs being easily transported by trucks or railways to the nuclear power site. Output, which is about one-third of traditional nuclear power reactors, may be scaled by combining several units to achieve larger capacities. So far, private sector involvement in the sensitive area of atomic energy had been limited to the construction of reactor vessels and the execution of other specific tasks as turnkey projects for government companies. The present collaboration with private players comes as India progresses on the path of achieving net-zero carbon emissions by 2070, requiring a transition to the use of cleaner fuels.

Meanwhile, the research boost for the space sector, as outlined in the current budget, comes in the footsteps of India's recent success in space exploration, including the successful Chandrayaan-3 and Aditya L1 missions, both accomplished last year. Industry analysts are optimistic about exponential growth in India's space market in a decade, with investments being made in launch vehicles, satellite manufacturing, Earth observation, communications and in-orbit economy. "These measures are crucial for the growth and development of our space ecosystem," says Lt. Gen. A.K. Bhatt (retd.), director general, Indian Space Association. Decoding the budget allocation, Srinath Ravichandran, CEO, Agnikul Cosmos, an aerospace manufacturer, says it "shows that the government is continuing to back its vision of making India have a larger chunk of the global space economy". According to government estimates, India's present installed nuclear capacity of 8,180 MW (megawatts) is slated to increase to 22,480 MW by 2031-32. In about the same period (by 2033), the country's space economy is likely to grow to $44 billion from the current 8.4 billion. The move to increase the involvement of private players in this journey, it is hoped, will accelerate the transition. ■

HIGHLIGHTS

- **Centre to partner with private entities to develop small modular reactors (SMRs) to boost nuclear energy production**
- **Exclusive venture capital fund for space startups**